# Earvin Tio

Machine Learning Engineer | Data Scientist

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## EXPERIENCE

**Research Assistant** | Vision and Image Processing Research Group
Supervised by Dr. Alexander Wong | Sept. 2019 - Apr. 2020

- Compared correlated diffusion imaging (CDI) and apparent diffusion coefficient (ADC) imaging for prostate cancer detection.
- Converted legacy Matlab code into user-friendly Python code. Presented a poster at the Conference on Vision and Intelligent Systems (CVIS) on an accepted submission.

**Bioinformatics Research Student** | SickKids Hospital (PGCRL)
Supervised by Dr. Adam Shlien | Sept. 2018 - Dec. 2018

- Compared gene expression quantification tools and machine learning algorithms used for tumor type prediction.
- Optimized bioinformatic pipelines to reduce runtime.
- Developed a novel pipeline (WDL, Bash, Python) that combines gene fusion detection and tumour type prediction using RNA-Seq data.

**Bioengineering Research Assistant** | Sunnybrook Hospital (SRI)
Supervised by Dr. Cari Whyne | Jan. 2018 - Apr. 2018

- Developed a Python module for 3D Slicer that fuses CT scans and ultrasound imaging used for navigation during spinal surgery.
- Implemented computer vision algorithms to track surgical tools.
- Refined a Python module for 3D Slicer that virtually simulates pedicle screw placement and grades performance; used for surgical training.
- Submitted an accepted abstract to the University of Toronto's Department of Surgery's 2018 SpineFEST conference.

**Research Assistant** | Waterloo Engineering Bionics Lab
Supervised by Dr. Ning Jiang | Sept. 2017 - Dec. 2017

- Ran experiments to collect surface electromyography (sEMG) data.
- Applied supervised machine learning algorithms in MATLAB to time series sEMG data for gesture recognition in biometric authentication.

**Medical Device Design Engineer** | Intellijoint Surgical Inc.
May 2017 - Aug. 2017

- Designed and prototyped the software (Swift, Objective-C) and hardware for navigated acetabular reaming, a highly surgeon-requested feature.
- Performed image processing (segmentation, thresholding, binary image operations) on computed tomography (CT) volumes in 3D Slicer to produce 3D-printed phantoms.
- Supported surgical cases by processing and visualizing data in MATLAB into client-friendly reports and intuitive statistics.

## EDUCATION

**Institute of Medical Science** MSc.
Collaborative Program in Neuroscience
University of Toronto | 2022

**Systems Design Engineering** BASc.
Option in Artificial Intelligence
University of Waterloo | 2020

**Computer Science** Exchange Semester
EPFL Switzerland | Feb. 2019 - June 2019

## SKILLS

**Machine Learning** | deep learning, neural networks, natural language processing

**Computer Vision** | image processing, object tracking, computer graphics

**Bioinformatics** | predictive statistics, differential expression analysis, gene quantification

## TOOLS

**Python** | Jupyter notebook, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib

**Version Control** | Git, GitHub, Atlassian (BitBucket, Confluence)

**Front End Development** | HTML5, CSS3, JavaScript, jQuery, JSON, XML

**Others** | MATLAB, R, LaTeX, C++, Bash

## PROJECTS

**Ambrosia** | machine learning driven mobile app that facilitates adoption of Intuitive Eating

**CaribbeanWAVE** | Teensy and Arduino driven implementation of MIDI on a redesigned steelpan

**Anarchy!** | 3D-printed and laser-cut original sociopolitical board game

**WaterMod** | 3D-printed water filtration and storage solution for third-world countries